$$F_{Ey} = k\frac{Qq}{r^2}\sin\theta = k\frac{Qq}{d^2}\cos^2\theta\sin\theta \ ,$$

Here $\theta$ is the angle of P, A line to x. The equation gives P at $\theta = 0 \ , \ \pi/2$, $F_{Ey} = 0$. On $0 \leqslant \theta \leqslant \pi/2$, as $\theta$ increases, $\sin\theta$ increases and $\cos^2\theta$ decreases. Thus on this interval $F_{Ey}$ has a maximum $F_{Ey\max}$; its angle $\theta_0$ can be found numerically. The calculated values are listed in Table 2.

Table 2

| $\theta$ / rad | 0.010 | 0.464 | 0.540 | 0.576 | 0.611 | 0.615 | 0.617 | 0.644 | 0.675 |
|---|---|---|---|---|---|---|---|---|---|
| $\cos^2\theta\sin\theta$ | 0.029 | 0.367 | 0.378 | 0.383 | 0.385 | 0.385 | 0.385 | 0.384 | 0.381 |

The table gives $\theta = \theta_0 \approx 0.615$ rad (i.e. 35.26°) at the maximum of $F_{Ey}$:

$$F_{Ey\max} = k\frac{Qq}{d^2}\cos^2\theta_0\sin\theta_0 = 0.385\,k\frac{Qq}{d^2} \ .$$

Charged particle P also feels gravity $G$, opposite to $F_{Ey}$. Hence P has a net force

$$F = F_{Ey} - G = k\frac{Qq}{d^2}\cos^2\theta\sin\theta - mg \ .$$

When $F = 0$, or $F_{Ey} = G$, P is in equilibrium. Thus if the mass satisfies

$$m \leqslant \frac{F_{Ey\max}}{g} = \frac{k\,(qQ/d^2)\cos^2\theta_0\sin\theta_0}{g}$$

, an equilibrium point exists on the $y$ axis.

6.

1. Imaging by refraction at one spherical surface obeys

$$\frac{n'}{s'} + \frac{n}{s} = \frac{n' - n}{r} \ , \qquad (1)$$

Here $s$, $s'$: object and image distances; $r$: radius; $n$, $n'$: refractive indices of the incident and refracted media, respectively.

Here object point P undergoes left-surface refraction (first image), then right-surface reflection (second), and left-surface refraction (third).

(1) First: $s_1$, $s'_1$: object and image distances. $s_1 = s$ , $n = n_0 = 1$ , $n' = n_g$ , $r = R$, we have

$$\frac{n_g}{s'_1} + \frac{1}{s_1} = \frac{n_g - 1}{R} \ , \qquad (2)$$